Externalized Courier – hospital version

November 2022, Noa Herz

Record only experiment

General structure (relevant for session 0 only)

1. Practice:
    a. Learning phase: Free navigation to 8 stores in the city.
    b. Standard FR: Deliver items to the remaining 8 stores (deliver 8 items, with the trial ending upon arrival at the 9th store). Then -> complete 10 sec of rest -> standard free-recall of items.
2. Response-required cued-recall phase (using a key press if guessing an item).
3. EFR practice delivery day, with feedback by the experimenter if needed. Deliver 12 items-> rest of 10 sec-> EFR (using a key press after incorrect recalls).
4. Continue the same structure of delivery days (deliver 12 items, rest of 10 sec, EFR and then response -required cued recall).
5. At the end of each session, complete externalized final free recall of all stores and objects seen on the session (using a key press after incorrect recalls).

General structure (relevant for session 1 onward)

1. Recap of instructions
2. Deliver 12 items-> rest of 10 sec-> EFR and then -> response -required cued recall.
3. At the end of each session, complete externalized final free recall of all stores and objects seen on the session (using a key press after incorrect recalls).

Full Methods

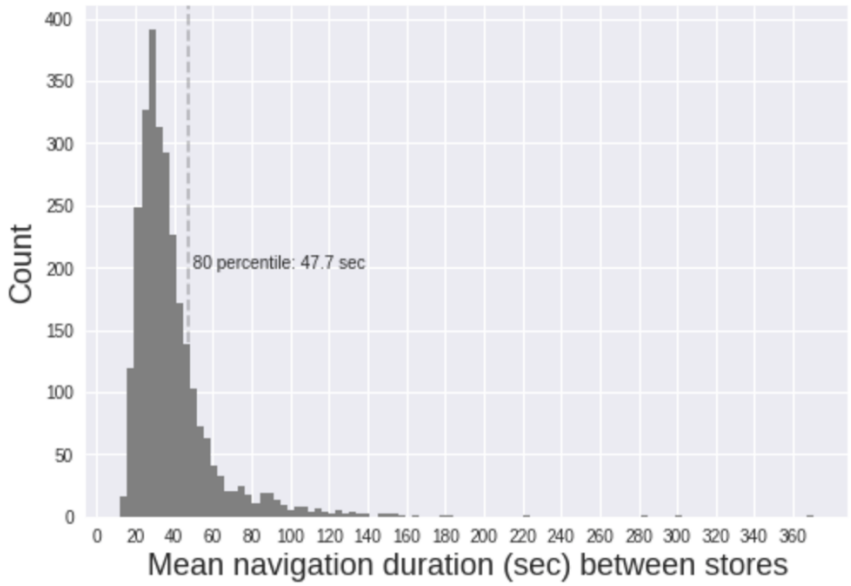
This version of Courier uses the same virtual navigation memory task as the one used in Delivery Boy, built using the Unity game engine software (Unity Systems). The below documentation focuses on changes made from the previous version. If any details are missing, assume they remained unchanged from the preceding hospital version.

In this navigation task, subjects are immersed in a small virtual town comprised of 17 labeled stores, as well as non-store buildings. The stores contain large banners displaying their names so that they can be identified from a distance, and the town also contains props such as fences, trees, mailboxes, street lights, and clouds. Subjects can move freely through the environment using a gamepad, and the task is completed on a testing laptop.

At the very beginning of the task, subjects are given the chance to learn the layout of the town by navigating sequentially to stores presented on the computer screen. Text instructing the subject where to go next is presented at the top-left corner of the screen during their navigation (i.e. “Please find the Bakery”). Upon arrival at the target store, accomplished by driving into a green circle located next to the store, ‘sparks’ appear and subjects are told which store to find next. They are then asked to point in the direction of the next store by adjusting an arrow in the straight direction of the required store and receive feedback about their response. After pointing the arrow, the arrow will change color on a gradient from green to red, according to the distance between the pointing and the real location. For answers within 30° from the correct store location (from both sides, meaning a total of 60°) glitters appear when choosing the location and a ‘good job’ text to indicate a correct response. For answers above 30° from the correct store location no glitters appear with a ‘not quite’ text. At the end of each delivery day (and not during navigation), a feedback appears telling participants the number of stores (out of the 13 stores they pointed to) they correctly pointed to (correct responses are those within 30° from the correct location). During this learning phase, subjects visit each of 8 stores once.

Then, subjects are asked to continue navigating while also recalling the items they deliver to the stores. Text will instruct subjects to navigate to the remaining 9 stores in the environment, to which they deliver 8 items. Upon arrival at each of the first 8 stores, the subjects are presented with an item semantically associated with the store (for example, "bread" may be presented at the bakery). Words do not repeat throughout delivery days of each session (i.e. each word appears once). Each item is presented on the screen both visually, for 1600 ms, and aurally, as vocal recordings from a single professional voice actor. Visual presentation of the words is done using the following parameters: Font Type: Arial, Background Color: transparent, word color: white. Words are presented ~1.25 sec after arrival at the green zone, with a jitter of +-0.250 sec (e.g. after 1.25 sec +-250 ms). Until words are presented movement is temporarily prevented.

If subjects navigate in the environment for 48 sec without arriving at the next store, an arrow will guide them in the direction of the next store to minimize their navigation time. The duration until providing path directions was determined based on retrospective analysis (n=40) showing that in 80% of trials subjects arrived at the next store within 48 sec:
Distribution of navigation times between stores (n=40)



Upon arrival at the 9th store, the screen turns black with a white fixation cross at the middle of the screen for 10 sec. The subjects are requested to look at the fixation cross during this rest period (only during the first delivery day of session 0, a text appears above the fixation cross, guiding participants: 'please look at the plus sign'). Then, a row of asterisks appears on the screen with a beep sound, signaling subjects to start recalling as many items as they can remember, in any order, from the just-completed practice day (using the standard FR instructions). They then complete a response-required cued-recall phase, in which each of the 8th stores to which they delivered items are presented on the screen, sequentially, and subjects are requested to recall the item delivered to each of the stores and guess an item in case they cannot remember what they delivered to that store. In these cases, patients are required to press the B button immediately after saying the item (only one item should be recalled per store). Each store appears on the screen with a text reminding subjects to press the B key after making a guess. After 7 sec the trial ends and the next store automatically appears. A circular progress bar at the lower right side of the screen shows participants how much time is left per trial.

During each subsequent delivery day subjects participate in a series of delivery days, each of which can be considered an equivalent of a list in the free recall (FR) tasks. The basic structure of the task remains the same as in the first delivery day of session 0, just without the video instructions, without the town

familiarization and without the standard free-recall phases. However, instead of the standard FR, subject now receive the instruction of the externalized free-recall:

The recall period after each delivery day is broken down into externalized free recall (EFR) and response-required cued recall. In the first, EFR, phase, subjects have 90 seconds to recite as many of the 12 items from that delivery day as they can remember in any order. They are now also requested to say any other, salient, word that comes to their mind even if it is a repetition of an item already recalled or an item not delivered on the just-completed delivery day, and subsequently press a key to indicate that the item recalled was incorrect (see verbatim instructions below). The second, response-required cued recall phase, stays the same as in the first delivery day of session 0: subjects are cued one at a time with 12 stores and are required to recite the corresponding item delivered to the store. If they can't remember the item, they are requested to guess and subsequently press a key (B) indicating that the item recalled was a guess.

During both recall periods, a text will appear on the screen reminding subjects to press the 'B' button after recalling items that they identify to be incorrect. Upon pressing the B key, the '(B)' becomes bolded and the font size is enlarged by 2 points to provide subjects a confirmation that the key was indeed pressed.

On the first EFR delivery day on session 0, provide feedback to participants according to their performance. The feedback is currently provided by experimenters according to the patient's needs.

(Later on, we will evaluate whether we should implement an automatically feedback, that will be presented on the computer screen when the conditions are met:

If:

Participant makes more than 3 intrusions and/or repetitions in a succession (i.e. press the 'B' key more than three times in a row).

Provide the feedback:

"Although the presented words may make you think of other related words, you should not recall all associated words. Keep in mind that your main task is still to recall as many words as possible from the most recently presented list."

If:

Participant never presses the 'B' key during the first EFR recall phase (i.e. never mark an intrusion).

Provide the feedback:

"Please keep in mind that we want you to vocalize every specific, concrete item that comes to your mind, even if you have already recalled it, or if it was not presented on the most recent delivery day. In these cases, press the B key after recalling that word, but before recalling the next word.")

At the end of each session, subjects complete a final free-recall (FFR) of all items delivered throughout the delivery days of that session, and all stores presented throughout the session, using a similar B press to reject recalled items (with a similar text appearing on the screen reminding subjects to use the B key, and with similar feedback once the key is pressed). The duration for FFR of items is set to 3 minutes, as previous data from the hospital shows that 90% of recalls happened within the first 2.7 minutes. FFR for stores is set to 90 seconds.

Throughout all delivery days, the layout of the town and store locations remains exactly the same as during the learning phase. During each delivery day, subjects are sequentially prompted to navigate to a specific store. The next store to be visited is selected according to these criteria (these criteria apply also for session 0):

1. Stores that were not-visited by the player in the current delivery day.
2. Stores that are not located behind the subject (i.e., stores that are between angles <=90 or >=270 of the current head direction of the player).
3. Stores that are not currently in view.

From the stores corresponding to criteria 1 -3, stores are ranked according to their distance from the player's current position. Then, one of the 3 stores that have the lowest distance from the player's current position are randomly chosen. If none of the remaining non-visited stores fall under criteria 2 and 3, then randomly select the next store until all of the 13 stores have been visited. These criteria for selecting the next store to be visited are meant to decrease patient's navigation time in the environment (therefore, reducing the time spent on each delivery day).

To reduce motion sickness, the following modifications of the previous hospital version were implemented (these were suggested by James Bruska):

- Environment lighting will change from daytime to night time (black skybox and reduced lighting)
- Increased physics rate from 30 to 120
- Fixed bicycle image on screen (no rotation animation)

List of main changes implemented in the current version relative to DBoy 3.0

	Delivery Boy 3.0	**Current version**
Display and gaming experience	• Day time • Bicycle image rotates when taking turns.	• Night time • Fixed bicycle image on screen (no rotation animation)
Practice	each of the 17 stores is flashed on the screen for 2 seconds, so that the subjects can learn how each store looks (outside the context of the actual town)	Stores familiarization was removed
	subjects visit each of the 17 stores thrice. However, if, after two rounds of visiting the stores, the experiment has taken longer than 45 minutes, then the third round of visits is not conducted, to minimize subject fatigue.	subjects visit each of 8 stores once. Then, they navigate to the remaining 9 stores while also delivering (8) items. Upon arrival at the 9th store, they conduct the standard FR and response required cued-recall phases.
	The first session is comprised of the store presentation and learning phase only	The first session is comprised of the learning phase and at least one EFR delivery day.
	No feedback is given during learning	Feedback is given, if necessary, during the first EFR delivery day on the first session.
Encoding	No assisting arrow	Following 48 sec without arriving at the next store, an arrow will guide subjects to the next store.
	After delivering each item, subjects will be required to point in the direction of the next store and will receive feedback about their	After delivering each item, subjects will be required to point in the direction of the next store and will not receiving any subsequent

	distance from the actual direction. Then, the arrow shows them the correct way to the next store.	guidance about the real location of the next store.
	Items during navigation are presented aurally	Items during navigation are presented both aurally and visually
	Subjects navigate to stores that are selected in a randomly determined order (without a repetition).	Subjects navigate to stores (without repetition) selected according to these criteria: 1. Only stores that are not behind the subject (i.e., stores that are between angles <=90 or >=270 from the current head direction). 2. Only stores that are not currently in view. 3. From the stores corresponding to criteria 1 and 2, randomly pick one of the 3 stores that have the lowest distance from current position. If none of the remaining non-visited stores apply to criteria 1 and 2, randomly select the next store until all of the stores have been visited.
Delay	No delay between navigation and the beginning of FR	A 10 sec rest between the end of navigation and the beginning of EFR
Recall phase	Standard FR	EFR with a key press for incorrect recalls
	Standard cued-recall: subjects are cued one at a time with 12 items or stores and given 5.5 seconds to recite the corresponding store or item	Response-requested cued-recall: subjects are cued one at a time with the 12 stores and are asked to recite the corresponding item or to guess an item if needed. Each trial will terminate 7 seconds after store presentation.
	Timing:	Timing:

	1. FR: 60 sec 2. Cued-recall: 5.5 sec 3. FFR of stores: 90 sec 4. FFR of objects: 5 minutes	1. EFR: 90 sec 2. Response-required cued-recall: 7 sec 3. FFR of stores: 90 sec. 4. FFR of objects: 3 min.

Stimulation experiment

The purpose of this experiment is to investigate how theta-burst stimulation during encoding and retrieval influence spatial and verbal episodic memory. The stimulation will be delivered while patients play the same externalized Courier task as outlined above.

Stimulation parameters
In this experiment, we will use:
f_{pulse} = {3,8} Hz.
The amplitude will be fixed to A=0.5ma:
With a fixed total duration (D) of 3 seconds.

Theta-burst duration, D_{burst}, will be fixed to:
D_{burst}= 0.25*(1/ f_{pulse}),
or 25% of the corresponding burst cycle. Since the total stimulation duration, D, will be fixed for 3 seconds, the number of bursts delivered will change according for the theta-burst frequency.

For f_{pulse} = 3 Hz, each stimulation cycle will last 1/3=0.333 seconds, with each theta-burst lasting 0.25*0.333=0.083 seconds. The total number of cycles will be 3/0.333=9.

For f_{pulse} = 8 Hz, each stimulation cycle will last 1/8=0.125 seconds, with each theta-burst lasting 0.25*0.125=0.031 seconds. The total number of cycles will be 3/0.125 =24.

Stimulation will be delivered using charge-balanced biphasic rectangular pulses (each phase of the pulse width = 300 μs, leading to a total of 600 μs pulse width), at 200 Hz.

Timing of stimulation

Encoding
Stimulation will be delivered for half of encoded items (6 stim words and 6 no-stim words). Selection of the stim/no-stim words will be done based on the stores in which they are encoded. Half of the existing 16 stores will be randomly allocated for a stim condition and half (i.e., 8) will be allocated to a no-stim condition. These stores will remain consistent within participants (i.e. the same 8 stores will always be stimulated for each participant across delivery days and sessions). Since each delivery day includes a total of 12 stores, half of these stores (i.e., 6) will be taken from the 8 stimulated pool and half from the no-stim pool.

For each stimulated item, theta-burst stimulation will last for 3 seconds, beginning from 1 second prior to word presentation and until 400 ms after the end of word presentation (words are presented for 1.6 seconds). A jitter of +-0.250ms for the 1 sec delay will be present before word presentation.

Stimulation frequency (3/8 Hz) will remain fixed within a list (=delivery day), but will alternate between lists, with the first list randomly allocated to either 3 or 8 Hz.

Retrieval (externalized free recall; EFR and final free-recall; FFR)
Stimulation will be delivered for 3 seconds on-off trains throughout the retrieval phase.
EFR for stores lasts 90 seconds, yielding 15 stim and 15 no-stim epochs.
FFR for items lasts 3 minutes, yielding 30 stim and 30 no-stim epochs.
The 1st 3 seconds epoch of the retrieval will always be non-stim and will then alternate between stim/no-stim until the end of the retrieval phase.

Theta-burst stimulation frequency (3/8 Hz) during EFR will be matched between encoding and retrieval (i.e., 3Hz stimulation during encoding will be followed by 3Hz stimulation during EFR, and 8Hz stimulation during encoding will be followed by 8Hz stimulation during EFR).

During FFR of stores and items, stimulation frequency will alternate between 8 and 3 Hz (with order counterbalanced across sessions and patients).

Brain regions
We would like to target LTC (similar location to the one used in Ezzyat Curr.Biol). In addition, we will prioritize MTL structures (with a specific prioritization of the entorhinal cortex).

Safe Amplitude Determination
Amplitude will be set to 0.5 mA. If after discharges are detected, amplitude will be lowered to 0.25 mA.

Going forward, we might want to consider changing the amplitude determination process, for example by moving to the same protocol as the one used for PS2:

Prior to the start of each experiment, we will determine a maximum current amplitude that does not cause after discharges (ADs). Specifically, for each frequency f^i_{pulse} in the set
$f_{pulse} = \{3,8\}$ Hz,

we will sweep the current amplitude from 0−1.5 mA in steps of 0.5 mA to find such maximum current amplitude A^i_{max}.

For each frequency-current combination, we will deliver a 1.5-second-long pulse with these parameters and wait for the neurophysiologist to check and report whether or not any ADs occurred. If ADs are observed for current amplitude A, the neurophysiologist will be asked to lower the current amplitude by 0.25 mA. If no ADs are reported at this lower value, A−0.25 mA current amplitude will be recorded as the maximum current amplitude. If ADs are observed, then A−0.5 mA will be recorded instead. If for some frequency f^i_{pulse} ADs are observed for the lowest value of the current, then that frequency value will not be tested in any of the subsequent experiments.
As a secondary check, once a maximum current amplitude A^i_{max} is established for the frequency f^i_{pulse}, we will deliver 5 trains of 1.5 s stimulation with a 2.75−3.25 s Inter-Stimulus Interval (ISI) at that amplitude and frequency to be certain that repeated stimulation with these parameters is safe.
For N = 2 different frequencies and a waiting time of roughly 10 seconds to check for ADs, the entire protocol for finding a safe value of the current should take about $2\times4\times(1.5 + 10) + 2\times5\times(1.5 + 3)\approx3$ minutes. The resultant set $A_{max}=\{A^1_{max}, A^2_{max}\}$ mA
of current amplitudes will be used in subsequent experiments.

Summary
We aim to collect <u>at least</u> 3 delivery-days with stimulation per patient (per session).
We will investigate the effects of stimulation on: proportion of correct recalls and false recalls (intrusions) in EFR and FFR, proportion of correct recall in the cued-recall phase, and memory for spatial information (navigation time and route). We will also test for differences between 3 and 8 Hz stimulation on all of these memory measures. Finally, in the FFR, we will test whether stimulation frequency functions as a 'contextual barcode' that cluster items according to their stimulation frequency (e.g. whether items stimulated with 3 Hz tend to cluster together during the FFR, or whether epochs stimulated with 3Hz during the FFR yield more recollection of items stimulated with 3Hz during encoding).